and leukoencephalopathy), the most common form of hereditary stroke disorder
- Vascular malformations: moyamoya disease. These children have abnormal vasculature. Moyamoya comes from the Japanese for 'puff of smoke', similar to the blurred appearance seen on angiography.

Investigations should include an assessment of cerebral vasculature with MRI and MRA and carotid Doppler studies to rule out dissection of the carotid arteries (formal angiography may be required later); echocardiography to detect a source of embolism, along with a thrombophilia and vasculitis screen, and metabolic tests for homocysteine and mitochondrial cytopathy. Often no cause can be identified. Rehabilitation requires the involvement of the remedial therapy team. Aspirin prophylaxis is recommended but further evidence is needed on the advisability of anti-thrombolytic agents.

Summary

Strokes
- Occur in infants and children
- Occur in the antenatal or perinatal period, and may present in late infancy with a hemiplegia or at the time with seizures
- Are most often seen in association with cardiac or sickle cell disease, but varicella infection is another cause.

Neural tube defects and hydrocephalus

Neural tube defects

Neural tube defects result from failure of normal fusion of the neural plate to form the neural tube during the first 28 days following conception. Their birth prevalence in the UK has fallen dramatically from 4 per 1000 live births in the 1970s to 0.15 per 1000 live births in 1998 and to 0.11 per 1000 live births in 2005 (Fig. 27.13). This is mainly because of a natural decline, as well as antenatal screening.

The reason for the natural decline is uncertain, but may be associated with improved maternal nutrition. Mothers of a fetus with a neural tube defect have a 10-fold increase in risk of having a second affected fetus. Folic acid supplementation reduces this risk. High doses are now recommended periconceptually for women with a previously affected infant planning a further pregnancy. Low-dose periconceptual folic acid supplementation is recommended for all pregnancies. In some countries e.g. United States, folic acid is added to flour for bread.

Anencephaly

This is failure of development of most of the cranium and brain. Affected infants are stillborn or die shortly after birth. It is detected on antenatal ultrasound screening and termination of pregnancy is usually performed.

Encephalocele

There is extrusion of brain and meninges through a midline skull defect, which can be corrected surgically. However, there are often underlying associated cerebral malformations.

Spina bifida occulta

This failure of fusion of the vertebral arch (Fig. 27.14a) is often an incidental finding on X-ray, but there may be an associated overlying skin lesion such as a tuft of hair, lipoma, birth mark or small dermal sinus, usually in the lumbar region. There may be underlying tethering of the cord (diastematomyelia), which, with growth, may cause neurological deficits of bladder function and lower limbs. The extent of the underlying lesion can be delineated using ultrasound and/or MRI scans. Neurosurgical relief of tethering is usually indicated.

Meningocele and myelomeningocele

Meningoceles (Fig. 27.14b) usually have a good prognosis following surgical repair.

Myelomeningoceles (Figs 27.14c, 27.15) may be associated with:
- Variable paralysis of the legs
- Muscle imbalance, which may cause dislocation of the hip and talipes
- Sensory loss
- Bladder denervation (neuropathic bladder)
- Bowel denervation (neuropathic bowel)
- Scoliosis
- Hydrocephalus from the Chiari malformation (herniation of the cerebellar tonsils and brainstem tissue through the foramen magnum), leading to disruption of CSF flow.

Management – The back lesion is usually closed soon after birth.

Paralysis and muscle imbalance – physiotherapy helps prevent joint contractures. Walking aids or a wheelchair help mobility.

For sensory loss – skin care is required to avoid the development of skin damage and ulcers.

Neuropathic bladder – an indwelling catheter may be required for a neurogenic bladder, or intermittent

Summary

Neural tube defects
- Include anencephaly, encephalocele, spina bifida occulta, meningocele and myelomeningocele
- The birth prevalence in the UK has fallen dramatically, mainly owing to a natural decline but also to antenatal screening
- The birth prevalence is reduced by periconceptual folic acid
- Myelomeningoceles can cause paralysis of the legs, dislocation of the hip and talipes, sensory loss, neuropathic bladder and bowel, scoliosis and hydrocephalus from the Chiari malformation.

Neural tube defects

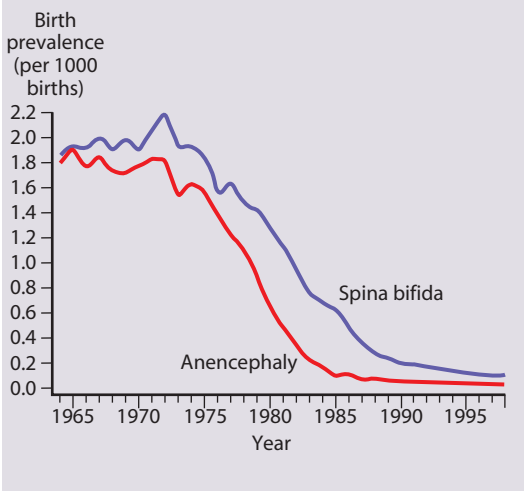


Figure 27.13 The decline in the number of babies born with neural tube defects. This has resulted from a natural decrease together with antenatal diagnosis and termination of pregnancy.



Figure 27.15 Myelomeningocele showing the exposed neural tissue and the patulous anus from neuropathic bowel.

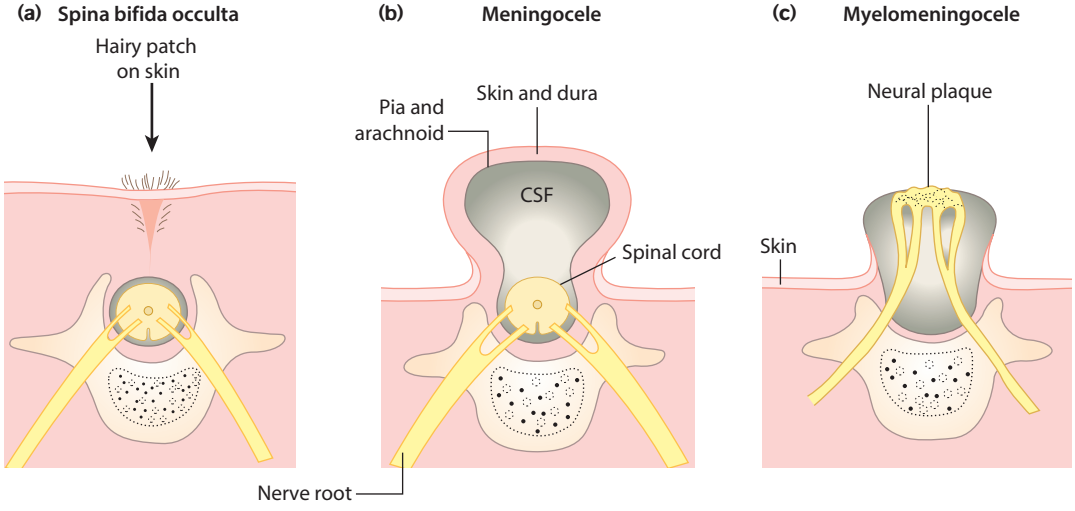


Figure 27.14 Neural tube defects: **(a)** spina bifida occulta; **(b)** meningocele; **(c)** myelomeningocele.

urinary catheterisation may be performed by parents or by older children themselves. There should be regular checks for hypertension, renal function and urinary infection. Prophylactic antibiotics may be necessary. Medication (such as ephedrine or oxybutynin) may improve bladder function and improve urinary dribbling.

Bowel denervation – requires regular toileting, and laxatives and suppositories are likely to be necessary with a low roughage diet for lesions above L3.

Scoliosis – is monitored and may require surgical treatment. Ventricular dilatation associated with a Chiari malformation – often present at birth and 80% of affected infants require a shunt for progressive hydrocephalus during the first few weeks of life.

The most severely disabled have a spinal lesion above L3 at birth. They are unable to walk, have a scoliosis, neuropathic bladder, hydronephrosis and frequently develop hydrocephalus.

Modern medical care has improved the quality of life for severely affected children. Their care is best managed by a specialist multidisciplinary team.

Hydrocephalus

In hydrocephalus, there is obstruction to the flow of cerebrospinal fluid, leading to dilatation of the ventricular system proximal to the site of obstruction. The obstruction may be within the ventricular system or aqueduct (non-communicating or obstructive hydrocephalus), or at the arachnoid villi, the site of absorption of CSF (communicating hydrocephalus) (Box 27.4).

Hydrocephalus



Figure 27.16 Grossly enlarged head and downward deviation of the eyes (setting-sun sign) from untreated hydrocephalus.

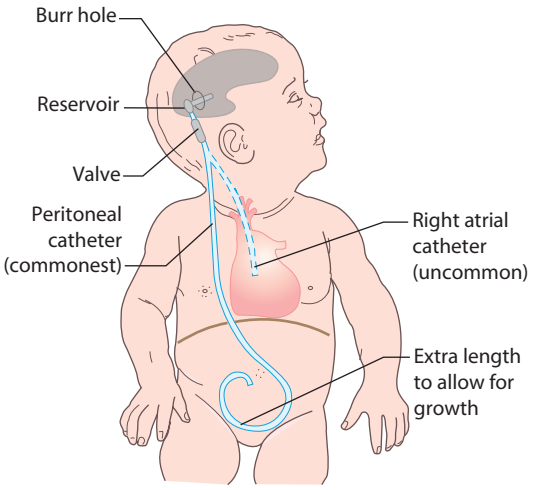


Figure 27.17 Ventriculoperitoneal shunt for drainage of symptomatic hydrocephalus. A sufficient length of shunt tubing is left in the peritoneal cavity to allow for the child's growth. Right atrial catheters require revision with growth.

Box 27.4 Causes of hydrocephalus

Non-communicating (obstruction in the ventricular system)

Congenital malformation
- Aqueduct stenosis
- Atresia of the outflow foramina of the fourth ventricle (Dandy–Walker malformation)
- Chiari malformation

Posterior fossa neoplasm or vascular malformation
Intraventricular haemorrhage in preterm infant

Communicating (failure to reabsorb CSF)

Subarachnoid haemorrhage
Meningitis, e.g. pneumococcal, tuberculous

Some can cause both non-communicating and communicating hydrocephalus.

Clinical features

In infants with hydrocephalus, as their skull sutures have not fused, the head circumference may be disproportionately large or show an excessive rate of growth. The skull sutures separate, the anterior fontanelle bulges and the scalp veins become distended. An advanced sign is fixed downward gaze or sun setting of the eyes (Fig. 27.16). Older children will develop signs and symptoms of raised intracranial pressure.

Hydrocephalus may be diagnosed on antenatal ultrasound screening or in preterm infants on routine cranial ultrasound scanning. For suspected hydrocephalus, initial assessment is with cranial ultrasound (in infants) or imaging with CT or MRI. Head circumference should be monitored over time on centile charts.

Treatment is required for symptomatic relief of raised intracranial pressure and to minimise the risk of neurological damage. The mainstay is the insertion of a ventriculoperitoneal shunt (Fig. 27.17), but endoscopic treatment to create a ventriculostomy can now be performed. Shunts can malfunction due to blockage or infection (usually with coagulase-negative staphylococci). They then need replacing or revising. Overdrainage of fluid can cause low-pressure headaches but the insertion of regulatory valves can help avoid this.

Summary

Hydrocephalus

- In infants, presents with excessive increase in head circumference, separation of skull sutures, bulging of the anterior fontanelle, distension of scalp veins and sun setting of the eyes
- Older children present with raised intracranial pressure
- Treatment is usually with a ventriculoperitoneal shunt.

The neurocutaneous syndromes

The nervous system and the skin have a common ectodermal origin. Embryological disruption causes syndromes involving abnormalities to both systems – the neurocutaneous syndromes.



Figure 27.18 Café-au-lait patches and axillary freckling in neurofibromatosis.



Figure 27.19 Adenoma sebaceum in tuberous sclerosis.



Figure 27.20 Sturge–Weber syndrome. There is a port-wine stain in the distribution of the trigeminal nerve.

Neurofibromatosis type 1 (NF1)

This affects 1 in 3000 live births. It is an autosomal dominant, highly penetrant condition. One-third have new mutations. The gene has been identified.

In order to make the diagnosis, two or more of these criteria need to be present:

- Six or more café-au-lait spots >5 mm in size before puberty, >15 mm after puberty (Fig. 27.18)
- More than one neurofibroma, an unsightly firm nodular overgrowth of any nerve
- Axillary freckles (Fig. 27.18)
- Optic glioma which may cause visual impairment
- One Lisch nodule, a hamartoma of the iris seen on slit-lamp examination
- Bony lesions from sphenoid dysplasia, which can cause eye protrusion
- A first-degree relative with NF1.

The cutaneous features tend to become more evident after puberty, and there is a wide spectrum of involvement from mild to severe. Neurofibromata appear in the course of any peripheral nerve, including cranial nerves. They may look unsightly or cause neurological signs if they occur at a site where a peripheral nerve passes through a bony foramen. Visual or auditory impairment may result if there is compression of the IInd or VIIIth cranial nerve. Megalencephaly with learning difficulties and epilepsy are sometimes seen.

Neurofibromatosis type 2 (NF2; bilateral, acoustic or central) is less common and presents in adolescence. Bilateral acoustic neuromata are the predominant feature and present with deafness and sometimes a cerebellopontine angle syndrome with a facial (VIIth) nerve paresis and cerebellar ataxia.

There may be an overlap between the features of NF1 and NF2. Both NF1 and NF2 can be associated with endocrinological disorders, the multiple endocrine neoplasia (MEN) syndromes.

Other associations are phaeochromocytoma, pulmonary hypertension, renal artery stenosis with hypertension, and gliomatous change, particularly in central nervous system lesions. Rarely, the benign tumours undergo sarcomatous change. However, most people with the disorder carry no features other than the cutaneous stigmata.

Tuberous sclerosis

This disorder is a dominantly inherited disorder, but up to 70% are new mutations. Prevalence is 1 in 9000 live births.

The cutaneous features consist of:

- Depigmented 'ash leaf'-shaped patches which fluoresce under ultraviolet light (Wood's light)
- Roughened patches of skin (shagreen patches) usually over the lumbar spine
- Adenoma sebaceum (angiofibromata) in a butterfly distribution over the bridge of the nose and cheeks, which are unusual before the age of 3 years (Fig. 27.19).

Neurological features are:

- Infantile spasms and developmental delay
- Epilepsy – often focal
- Intellectual impairment.

These children have severe learning difficulties and often have autistic features to their behaviour when older. Other features are:

- Fibromata beneath the nails (subungual fibromata)
- Dense white areas on the retina (phakomata) from local degeneration
- Rhabdomyomata of the heart which are identifiable in the early weeks on echocardiography but usually resolve in infancy
- Polycystic kidneys.

As with neurofibromatosis, gliomatous change can occur in the brain lesions. Many people who carry the gene have no stigmata other than the cutaneous features and no associated neurological features.

CT scans will detect the calcified subependymal nodules and tubers from the second year of life. MRI is more sensitive and more clearly identifies other tubers and lesions.

Sturge–Weber syndrome

This is a sporadic disorder with a haemangiomatous facial lesion (a port-wine stain) in the distribution of the trigeminal nerve associated with a similar lesion intracranially. The ophthalmic division of the trigeminal nerve is always involved (Fig. 27.20). Calcification of the gyri used to show characteristic 'rail-road track' calcification on skull X-ray, but MRI is the imaging modality of choice nowadays. In the most severe form, it may present with epilepsy, learning disability and hemiplegia. Children presenting with intractable epilepsy in early infancy may benefit from hemispherectomy. For children who are less severely affected, deterioration is unusual after the age of 5 years, although there may still be seizures and learning difficulties. There is a high risk of glaucoma, which should be assessed in the neonatal period.

Summary

The neurocutaneous syndromes

Include neurofibromatosis, tuberous sclerosis and Sturge–Weber syndrome.

Neurodegenerative disorders

These are disorders that cause a deterioration in motor and intellectual function. Abnormal neurological features develop, including seizures, spasticity, abnormal

Table 27.5 Lipid storage disorders

Disorder	Enzyme defect	Clinical features
Tay–Sachs disease	Hexosaminidase A	Autosomal recessive disorder Most common among Ashkenazi Jews Developmental regression in late infancy, exaggerated startle response to noise, visual inattention and social unresponsiveness Severe hypotonia, enlarging head Cherry red spot at the macula Death by 2–5 years Diagnosis – measurement of the specific enzyme activity Carrier detection of high-risk couples is practised Prenatal detection is possible
Gaucher disease	Beta-glucosidase	Occurs in 1 in 500 Ashkenazi Jews Chronic childhood form – splenomegaly, bone marrow suppression, bone involvement, normal IQ Splenectomy may alleviate hypersplenism Enzyme replacement therapy is available, but is expensive Acute infantile form – splenomegaly, neurological degeneration with seizures Carrier detection and prenatal diagnosis are possible
Niemann–Pick disease	Sphingomyelinase	At 3–4 months, feeding difficulties and failure to thrive, hepatosplenomegaly, developmental delay, hypotonia and deterioration of hearing and vision Cherry red spot in macula affects 50% Death by 4 years

head circumference (macro- or microcephaly), involuntary movement disorders, visual and hearing loss and behaviour change. While individually rare, they are numerous and include:

- Lysosomal storage disorders, e.g. lipid storage disorders and mucopolysaccharidoses, in which absence of an enzyme leads to accumulation of a harmful metabolite
- Peroxisomal enzyme defects, e.g. X-linked adrenoleucodystrophy. Peroxisomes are catalase- and oxidase-containing organelles involved in long-chain fatty acid oxidation. Enzyme deficiencies can lead to accumulation of very long-chain fatty acids (VLCFAs)
- Heredodegenerative disorders, e.g. Huntington disease, which presents with progressive dystonia, dementia, seizures and corticospinal tract signs
- Wilson disease, from the accumulation of copper, may cause changes in behaviour and additional involuntary movements or a mixture of neurological and hepatic symptoms
- Subacute sclerosing panencephalitis (SSPE), a delayed response in adolescence to previous measles infection causing neurological regression with a characteristic EEG, but has become rare since measles immunisation.

Lysosomal storage disorders

In metachromatic leucodystrophy, a sulfatidosis, an accumulation of sulphatides causes a destruction of myelin and is diagnosed on testing white cell enzymes.

In lipid storage disorders (Table 27.5), which are sphingolipidoses, there is an accumulation of

Mucopolysaccharidoses



Figure 27.21 Hurler syndrome showing the characteristic facies and skeletal dysplasia.

Table 27.6 Clinical features of mucopolysaccharidoses

Eyes	Corneal clouding
	Retinal degeneration
	Glaucoma
Skin	Thickened skin
	Coarse facies
Heart	Valvular lesions
	Cardiac failure
Neurology	Developmental regression
Skeletal	Thickened skull
	Broad ribs
	Claw hand
	Thoracic kyphosis
	Lumbar lordosis
Other	Hepatosplenomegaly
	Carpal tunnel syndrome
	Conductive deafness
	Umbilical and inguinal hernias

Table 27.7 Types of mucopolysaccharidoses

Type	Inheritance	Cornea	Heart	Brain	Skeletal
MPS I (Hurler)	AR	+++	++	+++	++
MPS II (Hunter)	X-linked	–	+	++	+
MPS III (Sanfilippo)	AR	±	–	+	+
MPS IV (Morquio)	AR	+	+	–	+++
MPS VI (Maroteaux–Lamy)	AR	+++	++	–	++

AR, autosomal recessive.

sphingolipids, essential components of CNS membranes. They are diagnosed on testing white cell enzymes.

The mucopolysaccharidoses are progressive multisystem disorders which may affect the neurological, ocular, cardiac and skeletal systems (Table 27.6). Hepatosplenomegaly is usually present. Most children present with developmental delay following a period of essentially normal growth and development up to 6–12 months of age. Developmental attainment then slows and children may show some loss of skills. It is only in the second 6 months of life that the characteristic facies begin to emerge, with coarsening of the facial features and prominent forehead due to frontal bossing (Fig. 27.21).

The characteristics of five of the varieties are shown in Table 27.7. The diagnosis is made by identifying the enzyme defect and the excretion in the urine of the major storage substances, the glycosaminoglycans (GAGs). Treatment is supportive according to the child's needs. Successful enzyme replacement by bone marrow transplantation has been performed but cannot reverse any established neurological abnormality.

Summary

In neurodegenerative disorders

There is developmental regression with the evolution of abnormal neurological signs.

Further reading

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Newton RW: *Colour Atlas of Pediatric Neurology*, London, 1995, Mosby-Wolfe.

A well-illustrated textbook.

Websites (Accessed May 2011)

British Paediatric Neurology Association: Available at: http://www.bpna.org.uk.

Child Neurology: Available at: http://www.familyvillage.wisc.edu/.

Provides 'family village' library with information on many neurological diagnoses.

International Headache Society classification: Available at: http://ihs-classification.org/en.

International League Against Epilepsy (ILAE): Available at: http://www.ilae.org.

Neuromuscular Disease Center: (Washington University School of Medicine, St Louis, MO). Available at: http://neuromuscular.wustl.edu.

Systematic reviews of migraine treatment in children, epilepsy, steroids for facial palsy and treatment of Guillain–Barré syndrome can be found in the Cochrane Library. Available via: http://www.thecochranelibrary.com.

28

Adolescent medicine

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Adolescence is the transition from childhood to adulthood. There is no clearly defined age range, but it is usually considered to be from puberty until 18 years of age. There are 7.8 million adolescents in the UK, 12–13% of the population, with increased proportions observed in ethnic minority groups.

The transition from being a child to an adult involves many biological, psychological and social changes (Table 28.1). Pubertal development is considered in Chapter 11. Difficulties may arise if the pubertal changes are early or delayed.

Table 28.1 Developmental changes of adolescence

	Biological	Psychological	Social
Early adolescence	Early puberty: Females – breast bud, pubic hair development, start of growth spurt Males – testicular enlargement, start of genital growth	Concrete thinking (Fig. 28.1a), but begin to develop moral concepts and awareness of their sexual identity	The early emotional separation from parents, start of a strong peer identification, early exploratory behaviours, e.g. may start smoking
Mid-adolescence	Females – end of growth spurt, menarche, change in body shape Males – sperm production, voice breaks, start of growth spurt Acne Blushing Need for more sleep	Abstract thinking, but still seen as 'bulletproof', increasing verbal dexterity, may develop a fervent ideology (religious, political)	Continuing emotional separation from parents, heterosexual peer interest, early vocational plans
Late adolescence	Males – end of puberty, continued growth in height, strength and body hair	Complex abstract thinking (Fig. 28.1b), identification of difference between law and morality, increased impulse control, further development of personal identity, further development or rejection of ideologies	Social autonomy, may develop intimate relationships, further education or employment, may begin or develop financial independence

Adapted from Adolescent development. In: Viner R, ed. 2005. *ABC of Adolescence*. Blackwell, Oxford.

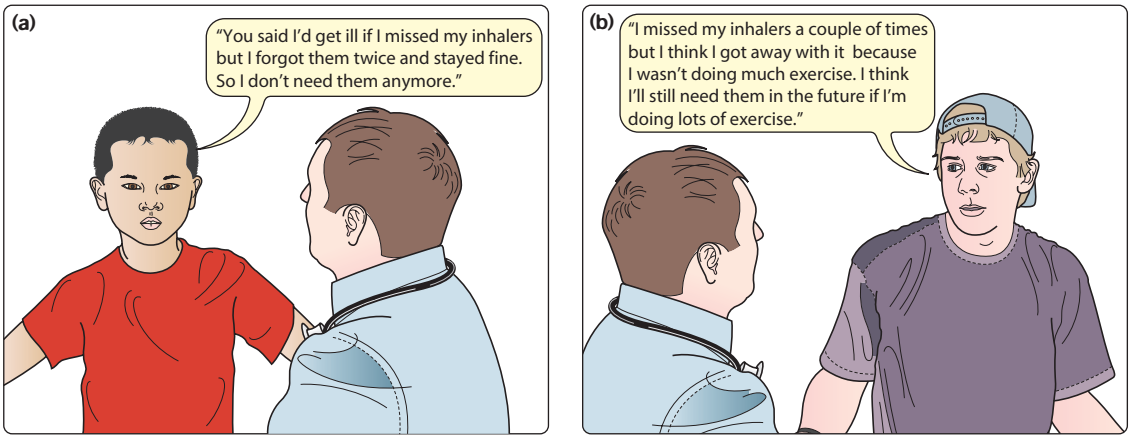


Figure 28.1 Example showing the difference between **(a)** concrete and **(b)** abstract thinking in the management of asthma in an older child and an adolescent.

While general practitioners will see all adolescent medical problems, difficulties may arise when obtaining specialist medical care. Those less than 16 years old are generally looked after by paediatricians; over 16 years old, by either paediatricians or more often by adult physicians and surgeons. However, paediatric facilities, e.g. children's wards, are often geared to the needs of young children rather than adolescents, whilst older adolescents may be overwhelmed by the medical conditions encountered on adult wards and the independence expected of them. Adolescent females with gynaecological problems are often cared for by gynaecologists, usually in adult facilities. Some paediatricians in the UK are now specialising in adolescent medicine in a similar way to North America and Australia.

Communicating with adolescents

The adolescent consultation differs from the paediatric consultation for young children, in that the adolescent has a greater active role in the consultation.

As well as seeing adolescents with their parents, an integral component of adolescent healthcare is offering young people the opportunity to be seen independently of their parents for at least part of the visit The principle is that the parents should not be seen alone after the adolescent has spent time with the doctor, so that the adolescent can trust that whatever confidences have been disclosed to the doctor have been kept.

Some practical points about communicating and working with adolescents are:

- Make the adolescent the central person in the consultation.
- Be yourself. When establishing rapport, it may be appropriate to engage the adolescent by talking about their interests, e.g. football, clothes or music, but do not try to be cool, false or patronising; your relationship should be as their doctor, not their friend.
- Consider the family dynamics. Is the mother or father answering for the adolescent? Does the adolescent seem to want this or resent being interrupted?
- Avoid being judgemental or lecturing. Avoid 'You ….' statements and use 'I ….' statements in preference, e.g. 'I am concerned that you ….'. A frank and direct approach works best. Your role should be that of a knowledgeable, trusted adult from whom they can get advice if they so choose.
- An authoritarian approach is likely to result in a rebellious stance. Working things out together in a practical way has the best chance of success.
- Frame difficult questions so they are less threatening and judgemental, e.g. 'Lots of teenagers drink alcohol, do any of your friends drink? How much do they drink in a week? Do you drink alcohol – how much do you drink compared to them?' Likewise, when asking sensitive questions on, e.g. sexual health, always give young people warning and explain the rationale of why such questions need to be asked.
- Confidentiality is particularly important to this age group and must be respected. Explain that you will keep everything you are told confidential, unless they or somebody else is at risk of serious harm. Always assess their understanding of confidentiality and correct any misunderstanding.
- Bear in mind proxy presentations, e.g. abdominal pain, when the real reason is anxiety about the possibility of pregnancy, or sexually transmitted infection or the result of recreational drug use.
- A full adolescent psychosocial history is useful to engage the young person, to assess the level of risk as well as provide information which will aid the formulation of effective interventions. The HEADS acronym may be helpful in this regard (Table 28.2), although questions must always be tailored to stage of development and the right of the young person to not answer should be respected.

Table 28.2 HEADS acronym for psychosocial history taking in adolescents

H	**Home life**	Relationships, social support, household chores
E	**Education**	School, exams, work experience, career, university, financial issues
A	**Activities**	Exercise, sport, other leisure activities
		Social relationships, friends, peers, who can they rely on?
D	**Driving**	Aged 16 if has high rate mobility component of the Disability Living Allowance (DLA)
	Drugs	Drug use, cigarettes, alcohol. How much? How often?
	Diet	Weight, caffeine (diet drinks), binges/vomits
S	**Sex**	Concerns, periods, contraception (and in relation to medication)
	Sleep	How much? Hard to get to sleep? Wake often?
	Suicide/affect	Early waking? Depression, self-harm, body image

- Communicate and explain concepts appropriate to their cognitive development. For young adolescents, use concrete examples ('here and now') rather than abstract concepts ('if … then') .
- History-taking should avoid making the assumption of heterosexuality with questions about romantic and sexual partners asked in a gender neutral way.
- If they need to have a physical examination, consider their privacy and personal integrity – Who do they want present? As with any age, young people have the right to a chaperone but it should not be assumed the young person will want this to be their parent. Also, find out if they would prefer a doctor of the same sex, if this is an option.

Summary

Talking and listening with young people

- Always give them the opportunity to be seen independently of their parents
- Explain and assure confidentiality
- Psychosocial screening is useful to:
  - Engage young people
  - Assess risk
  - Assist formulation of interventions.

Consent and confidentiality

Consent

In the UK, young people can give consent if they are sufficiently informed and either over 16 years old or under 16 years and competent to make decisions for themselves. Conflict rarely arises about a treatment, as usually the adolescent, their parents and doctors agree that it is necessary. Handling of disagreement over consent is considered in Chapter 5.

Confidentiality

Confidentiality is regarded by adolescents as of crucial importance in their medical care. They want to know that information they have disclosed to their doctor is not revealed to others, whether parents, school or police, without their permission. In most circumstances, their confidentiality should be kept unless there is a risk of serious harm, either to themselves from physical or sexual abuse or from suicidal thoughts or to others from homicidal intent. Difficulties relating to confidentiality for adolescents are usually about contraception, abortion, sexually transmitted infections, substance abuse or mental health. It is usually desirable for the parents to be informed and involved in the management of these situations and the adolescent should be encouraged to tell them or allow the doctor to do so. However, if the young person is competent to make these decisions for himself/herself, the courts have supported medical management of these situations without parental knowledge or consent.

Range of health problems

Adolescence is considered a healthy stage of life compared with early childhood or old age. In spite of this, the majority of young people will consult their general practitioner more than once in a year and 13% of adolescents report a chronic illness. The range of health problems affecting adolescents include:

- Common acute illnesses: respiratory disorders, skin conditions, musculoskeletal problems including sports injuries and somatic complaints. Acute serious illness has become rare, with mortality predominantly from trauma
- Chronic illness and disability: e.g. asthma, epilepsy, diabetes, cerebral palsy, juvenile idiopathic arthritis, sickle cell disease. The prevalence of some of the common chronic disorders in adolescence is shown in Table 28.3. There is also a range of uncommon disorders with serious chronic morbidity such as malignant disease and connective tissue disorders. In addition, children

Table 28.3 Prevalence of some chronic illnesses per 1000 adolescents (12–18 years old)

Disease	Prevalence per 1000 adolescents
Musculoskeletal conditions	41
Skin conditions	32
Significant mental health problems	120
Diabetes	
Type 1	2
Type 2	1–2
Respiratory conditions	150
Asthma	100
Cystic fibrosis	0.1
Epilepsy	4
Hearing problems	18
Cerebral palsy	1.5

with many congenital disorders which often used to be fatal in childhood now survive into adolescence or adult life, e.g. cystic fibrosis, Duchenne muscular dystrophy, complex congenital heart disease, metabolic disorders, etc.
- High prevalence of somatic symptoms: fatigue, headaches, backache, etc.
- Mental health problems including suicide and deliberate self-harm
- Eating disorders and weight problems
- Those associated with health-risk behaviours, such as smoking, drinking, drug abuse and sexual health, contraception and teenage pregnancy.

Mortality

The dramatic improvement in the mortality of young children seen since the 1960s has not been matched in adolescents, who now have a higher mortality rate than that of 1–4-year-olds (Fig. 28.2). Although deaths in adolescents from communicable diseases have declined markedly, this has not been matched by mortality from road traffic accidents, other injuries and suicide, and these now predominate (Fig. 28.3). Alcohol is thought to be a contributing factor in one-third of these deaths.

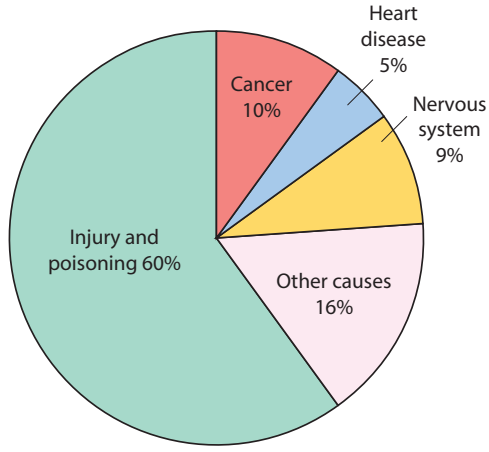


Figure 28.3 Causes of death, 15–19 years of age, in England and Wales, 2008. (Source: ONS, 2010.)

Impact of chronic conditions

Chronic illness may disrupt biological, psychological and social development. In addition, these developmental changes may affect the control and management of the disorder (Table 28.4). The impact of chronic illness on children, young people and their families is considered in Chapter 23.

Adherence

Poor adherence is a problem for many people, including adolescents as they are beginning to take over management of their illness, wish to avoid parental supervision and may give the management of their illness a lower priority than social and recreational activities. They may not believe that taking the

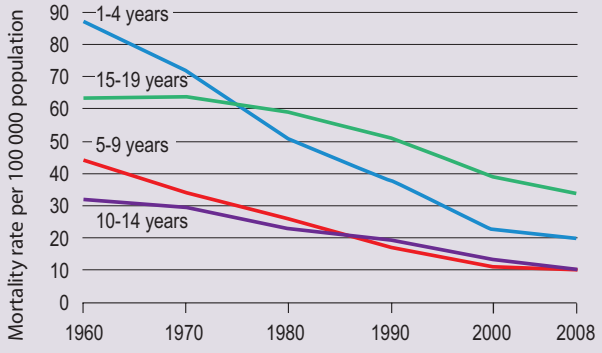


Figure 28.2 Mortality by age group in England and Wales 1960–2008, showing that the mortality rate is now greater at 15–19 years than at 1–4 years. (Source: ONS, 2010.)

Table 28.4 Some of the ways in which chronic illness and development interact with each other

	Effect of chronic illness on development	Effect of development on chronic illness
Biological	Delayed puberty Short stature Reduced bone mass accretion Malnutrition secondary to inadequate intake due to increased caloric requirement of disease or anorexia Localised growth abnormalities in inflammatory joint disease, e.g. premature fusion of epiphyses	Pubertal hormones may impact on disease, e.g. growth hormone worsens diabetes and increases insulin requirements; females with cystic fibrosis may have deterioration in lung function; corticosteroid toxicity worse in peripubertal phase Increased caloric requirement may worsen disease control or result in undernutrition – may need dietary supplements or overnight feeding with nasogastric tube or gastrostomy Growth may cause scoliosis
Psychological	Regression to less mature behaviour Adopt sick role Impaired development of sense of attractive/sexual self Parental stress, depression, financial problems in providing care; siblings may suffer	Deny that their health may suffer from their actions Poor adherence and disease control Reject medics like parents
Social	Reduced independence when should be separating Failure of peer relationships Social isolation – unable to participate in sports or social events School absence and decline in school performance, may lower self-esteem Vocational failure	Risk behaviour may adversely affect disease, e.g. smoking and asthma or cystic fibrosis, alcohol and diabetic control, sleep deprivation and epilepsy Chaotic eating habits lead to malnutrition or obesity

medication really matters, especially if it is preventative or of long-term rather than short-term benefit.

Peer relationships and self-image are very important when considering adherence. For example, it may be more important for an adolescent with diabetes to lunch promptly, so he can sit with his friends rather than go to the school nurse first for his insulin injection. Side-effects are also important, particularly those that affect well-being or appearance. They may assess risk differently from adults, so that the risk of not being one of their crowd because of having to adhere to a certain treatment may appear to be more important than the risks attached to not taking any medication.

Adherence may be influenced by lack of knowledge and/or poor recall of previous disease education. The disorder may have presented when the child was much younger, so that the original consultation will have taken place primarily between the doctor and parents. If this communication has not been updated with increasing age, the adolescent's knowledge may be poor, with little understanding about his/her illness, what medications he/she is taking and why. As the responsibility for management moves to the young person, information needs to be provided about medications and treatment appropriate for his/her development. Other ways to maximise adherence are summarised in Table 28.5.

The implications of their condition on the rest of their health needs to be considered. This may include sexual health, future vocational development, including the need for disclosure and their rights under the Disability Discrimination Act. Similarly, the implications of other health-risk behaviours such as substance use, tattoos and piercing may need to be discussed.

Transition to adult services

The young person with a chronic condition must eventually leave paediatric and adolescent services for adult services. This often involves changing from a treatment model based around close contact between the adolescent and healthcare professionals (unlimited telephone advice from clinical nurse specialists, possibly home visits, frequent appointments) and involvement with parents and other family members, to one where they are likely to be seen infrequently in a busy adult

Table 28.5 Ways to maximise adherence

Assess the size of the problem and be non-judgemental	Ask: 'Most people have trouble taking their medication. When was the last time you forgot?'
Take time to explore practicalities	Try to put yourself in the adolescent's shoes and think through the detail of their regimen with them. Make regimen as simple as possible. Don't forget practical issues – poor adherence may be as simple as not having anywhere private at school to take the treatment
Explore beliefs	May harbour strange or incorrect beliefs about medications, e.g. falsely attribute a side-effect and therefore refuse to take the medication
Use daily routines to 'anchor' adherence	Find daily activities to anchor taking the medication, e.g. brushing teeth, or 'with breakfast and dinner' instead of 'twice a day'. Find the least chaotic time of day: may be morning or evening! Let the suggestions come from the adolescent
Motivation	Negotiate short-term treatment goals. Search for factors that motivate the young person
Involve and contract	Plan the regimen with the adolescent. Some may respond to a written contract that both sides agree to stick to
Written instructions	Most of what is said has been shown to be forgotten once they leave the room!
Take time to explain	Check level of knowledge on each occasion
Solution-focused approach	Find out what has been going well and why. Use this information, e.g. 'How have you managed to remain out of hospital for 3 weeks this month?'

clinic where parental involvement may be minimal or discouraged.

Young people and their parents need both information about the transfer process and time to prepare. Transitional care encompasses this preparation which, by definition, addresses the medical, psychosocial and educational/vocational needs as a young person moves from child- to adult-centred services. Parents are often concerned that the adult team will not address their teenager's healthcare needs. It is helpful if an identified healthcare professional, often a nurse specialist, is responsible for coordinating transition arrangements.

Whereas transitional care starts in early adolescence, some flexibility in age of transfer is desirable, so that it can occur when the young person is developmentally ready and has the necessary maturity to cope with adult services.

Transfer may be via an adolescent or young adult service with clinics run by both adolescent and adult teams together. Such bridging arrangements have many advantages, but require a sufficient number of patients and medical staff able and willing to provide this service. These clinics are usually for specialist conditions, e.g. diabetes, juvenile idiopathic arthritis, cystic fibrosis or congenital heart disease. Alternatively, transfer may be successfully accomplished if there is good communication between teams, although it usually involves a radical change in ethos for the adolescent and family. The general practitioner may be a source of continuity between changing specialty practitioners.

Summary

Chronic conditions during adolescence

- Chronic illness and/or disability may disrupt adolescent development
- Consideration should be made of the impact of the chronic condition on the rest of health (including sexual and reproductive health) as well as education and leisure
- Transitional care aims to address medical, psychosocial and educational/vocational issues as young people move from child- to adult-centred services.

Fatigue, headache and other somatic symptoms

Fatigue, headache, abdominal pain, backache and dizziness are common in adolescence. International surveys of adolescents in Europe reveal that two-thirds report morning fatigue more than once a week, 25% have a headache and 15% stomach ache, backache or sleep problems more than once a week. In many, these symptoms appear to be a feature of adolescence, although organic disease must be excluded by history, examination and, occasionally, investigation. For a minority, they may be a physical manifestation of

Table 28.6 Main mental health problems and disorders in adolescents

Problem or disorder	Prevalence (%)
Depression	3–5
Anxiety	4–6
Attention deficit hyperactivity disorder	2–4
Eating disorders	1–2
Conduct disorder	4–6
Substance misuse disorder	2–3

Source: Michaud P-A, Fombonne E. 2005. Common mental health problems. In: Viner R, ed. 2005. *ABC of Adolescence*. Blackwell, Oxford.

psychological problems, and are precipitated by or maintained by factors such as bullying or parental discord.

Occasionally, the symptoms are so severe and persistent that they considerably affect quality of life, with impairment of school attendance, academic results and peer relationships. This may be from chronic fatigue syndrome or chronic idiopathic pain syndromes. Further investigation and assessment will be required and multidisciplinary rehabilitation and cognitive behavioural therapy within the family may be beneficial. The management of somatic symptoms and chronic fatigue syndrome are considered further in Chapter 23.

Mental health problems

The prevalence of mental health problems in adolescents is estimated to be about 11%. The main problems are listed in Table 28.6.

Deliberate self-harm varies from little actual harm, where there is a wish to communicate distress or escape from an interpersonal crisis, to suicide. About 7–14% of adolescents will self-harm, depending on its definition.

Eating disorders are common during adolescence. About 40% of females and 25% of males begin dieting in adolescence because of dissatisfaction with their body. In anorexia nervosa and bulimia, there is a morbid preoccupation with weight and body shape. This is discussed in more detail in Chapter 23.

Health-risk behaviour

During adolescence, young people begin to explore 'adult' behaviours, including smoking, drinking, drug use and sex. These behaviours, often referred to as 'risk-taking' behaviours, may reflect the adolescent's search for pleasure and excitement by participating in new and enjoyable experiences, as well as exerting independence from parents or rebelling against parents' wishes and lifestyle. There is also considerable pressure to fit in with peers.

Adolescents do not always understand the risks involved and may behave as if they are immune from harm. Participating in these activities may also deflect attention away from themselves to mask shyness or anxiety. Unfortunately, health-risk behaviours started in adolescence tend to continue into adult life.

Sexual health

The average age for first sexual intercourse in the UK is 16 years, with one-fifth of 14-year-olds having had intercourse. Having sexual intercourse at an early age is often associated with unsafe sex. This may be because of a lack of knowledge, lack of access to contraception, inability to negotiate obtaining contraception, being drunk or high on drugs or unable to resist being pressurised by their partner.

Risk-taking behaviour in adolescents can result in sexually transmitted infections (STIs) or unplanned pregnancy. STIs may present with urethral or vaginal discharge, urinary symptoms, pain on micturition, abdominal or loin pain, or post-coital vaginal bleeding. Chlamydia is asymptomatic in 50% of cases and can lead to later infertility. In young teenagers, it is more likely to present with a vaginal discharge. Studies have shown that up to one-third of sexually active teenage girls have a sexually transmitted infection. They are also at risk of HIV infection.

Management of sexually transmitted infections

Taking a sexual history from an adolescent should be approached sensitively, in a developmentally appropriate manner, giving the young person warning of the topic, as well as why the questions are being asked. Relevant questions include those related to the risk of STIs: number of partners; any partners during travel abroad; contraception used; whether vaginal, oral or anal sex; any discharge, lower abdominal pain, urinary symptoms; last menstrual period. However, many sexually transmitted infections are asymptomatic, especially in younger teenagers, male and female.

If indicated, swabs should be taken for virology and microbiology (to look for human papillomavirus (HPV), herpes simplex virus (HSV), chlamydia and gonorrhoea). HIV testing may be indicated. In England, in response to the high rates of chlamydia in the under-25-year-old age group, there is a national chlamydia screening programme enabling them to test themselves with easy-to-use kits.

Treatment regimens vary, depending on prevalent antibiotic resistance. Chlamydia can be treated with azithromycin or doxycycline, gonorrhoea with a cephalosporin. Metronidazole can be added for pelvic inflammatory disease. It is advisable to inform and treat partners.

Contraception

Most adolescents who are sexually active *are* using contraception, albeit sometimes haphazardly. In the UK, contraception is used by only half at first intercourse. Condoms, followed by the oral contraceptive pill, are the commonest forms of contraception used. As teenagers have a relatively high failure rate in their ability to use condoms correctly and with the oral contraceptive pill having irregular use, the 'double Dutch' method of condom and oral contraception is advocated to protect against both sexually transmitted infections and pregnancy.

Adolescents with chronic disease, e.g. diabetes, even without microvascular complications, are generally started on lower doses of the contraceptive pill. Some medications prescribed in adolescents are potentially teratogenic (e.g. retinoids for acne, methotrexate for juvenile idiopathic arthritis or other disorders) and may therefore need to be combined with an oral contraceptive pill or depot hormonal implant. Discussions, however, must also reinforce condom use to prevent STIs.

Emergency contraception

Emergency contraception (in the past misleadingly known as the 'morning after pill') can provide significant protection from pregnancy for up to 72 h after unprotected intercourse. Emergency contraception is available from a pharmacist without prescription for those 16 years and over, and on prescription for those under 16 years. If taken within 72 h, it has a 2% failure rate. Side-effects include nausea and lethargy. However, knowledge of emergency contraception is poor among many young people.

Teenage pregnancy

The UK has the highest rate of teenage pregnancy in Western Europe. Teenage girls may present with complaints such as abdominal pain, fatigue, breast tenderness or appetite changes rather than late or missed menstrual period.

Becoming a teenage mother can be a positive life choice and is influenced by culture. There may be considerable support from the extended family, and this may work well. However, in those where the pregnancy was unintended or who are emotionally deprived themselves or unsupported and live in poverty, there may be many adverse consequences for the mother and child. Children of teenage mothers have a higher infant mortality, a higher rate of childhood accidents, illness and admission to hospital, being taken into care, low educational achievement, sexual abuse and mental health problems. Deprivation, from the mother's lack of financial and emotional support and the paucity of her own education and life experiences, is the strongest risk factor. Protective factors are having a supportive family, religious belief and a stable, long-term relationship with the partner.

Health promotion

The reasons to undertake health promotion in adolescents are:

- It is the period for starting health-risk behaviours (smoking, alcohol, drug misuse, unsafe sexual activity)
- Health-risk behaviours started in adolescence often continue into adult life
- Health behaviours may have a direct effect on their lives, e.g. teenage pregnancy, road traffic accidents
- Increasing morbidity, e.g. obesity and diabetes.

The main areas for health promotion are:

- Health-risk behaviours
- Mental health
- Violent behaviour
- Physical activity, nutrition and obesity
- Parent–adolescent communication.

There are a number of approaches to health promotion for adolescents:

- Provide suitable information in a user-friendly way for young people. An example is the website: Teenage Health Freak (Fig. 28.4).
- Health promotion by society as a whole, e.g. banning cigarette advertising, making emergency contraception available in pharmacies. These can be very effective. However, there is increasing evidence that improving the socioeconomic circumstances of young people would be the most effective intervention for health promotion. Also, as adolescents often embark on more than one risk behaviour, tackling the underlying problem may reduce other risk-taking behaviours: e.g. a programme to reduce bullying in a whole school may also reduce other behaviour such as drug misuse.
- Training programmes to improve adolescents' ability to accept or reject certain courses of behaviour can be effective for the individual, but is time-consuming and expensive.



Figure 28.4 The website Teenage Health Freak promotes health in a user-friendly way to teenagers. Available at: http://www.teenagehealthfreak.org (Accessed May 2011). (Reproduced with permission.)

Summary

The main health problems of adolescents



Common acute illnesses

Chronic illness and disability, including previously fatal congenital disorders

Somatic symptoms:
- Fatigue, headache, backache and abdominal pain (see Ch. 23)

Mental health problems (see Ch. 23)

Health-risk behaviours:
- Smoking, drinking, drug abuse, road traffic accidents (see Chs 1 and 23)

Sexual health:
- Sexually transmitted infections, contraception, teenage pregnancy

Eating disorders (see Ch. 23) **and obesity** (see Ch. 12)

- Health promotion by professionals. Exhorting adolescents not to smoke, to eat a balanced diet, use contraception, etc., has not been found to be effective, and may be counter-productive. Health professionals do have a role in health promotion at an individual level. It is likely to be most effective if targeted at those who are receptive or contemplating change in their health-risk behaviour. However, motivational interviewing techniques (which do not assume that they are ready to change their behaviour, but aim to increase their intrinsic motivation to change) have also been shown to be useful with this age group.

Further reading

Not Just a Phase: A Guide to the Participation of Children and Young People in Health Services, London, 2010, Royal College of Paediatrics and Child Health (RCPCH).

Viner R: *ABC of Adolescence*, London, 2005, BMJ Books.

Websites (Accessed May 2011)

Teenage Health Freak: Available at: http://www.teenagehealthfreak.org.

Youth Health Talk: Available at: http://www.youthhealthtalk.org.

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Appendix

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Growth charts

Examples of growth charts used in the UK for:

- low birthweight males (A.1a) and females (A.1b)
- 0–1 years, boys (A.1c) and girls (A.1d)
- 1–4 years, boys (A.1e) and girls (A.1f)
- Birth–20 years, boys (A.1g).

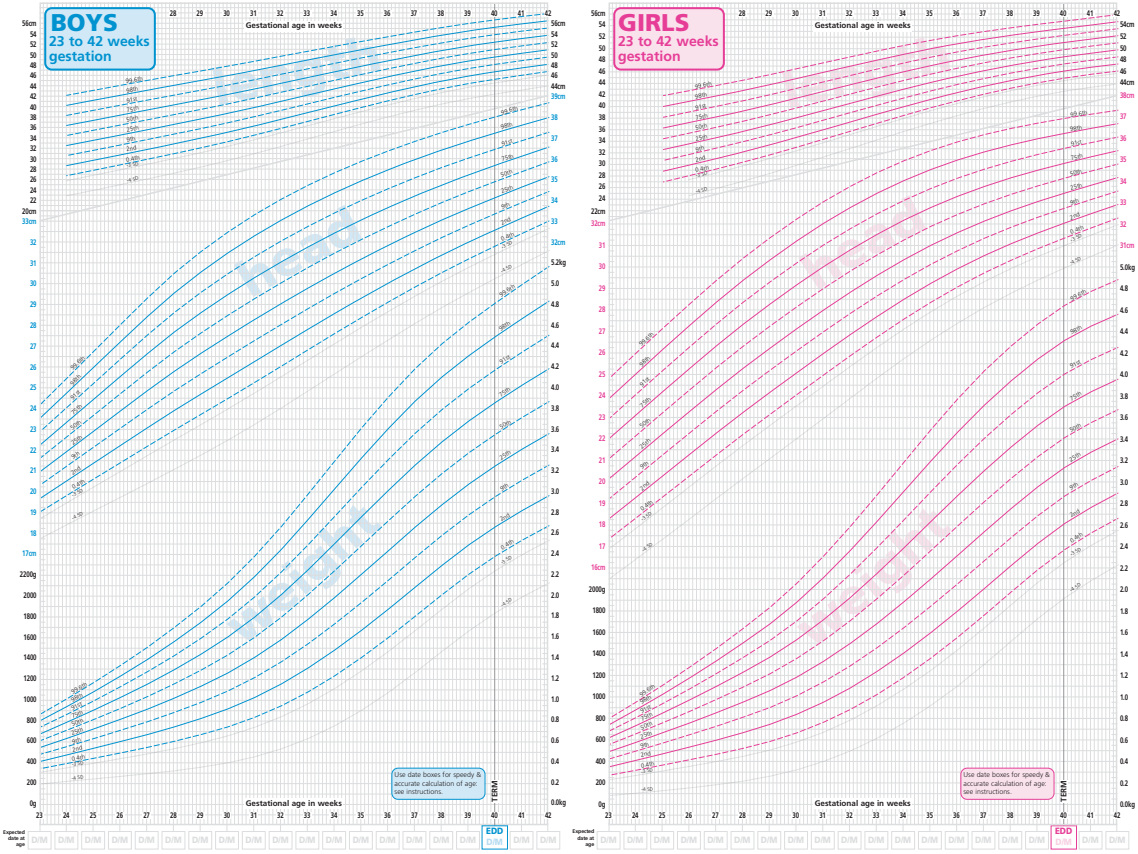


Figure A.1 Growth charts for low birthweight males (A.1a) and females (A.1b). (RCPCH/WHO/Department of Health. ©2009 Department of Health).

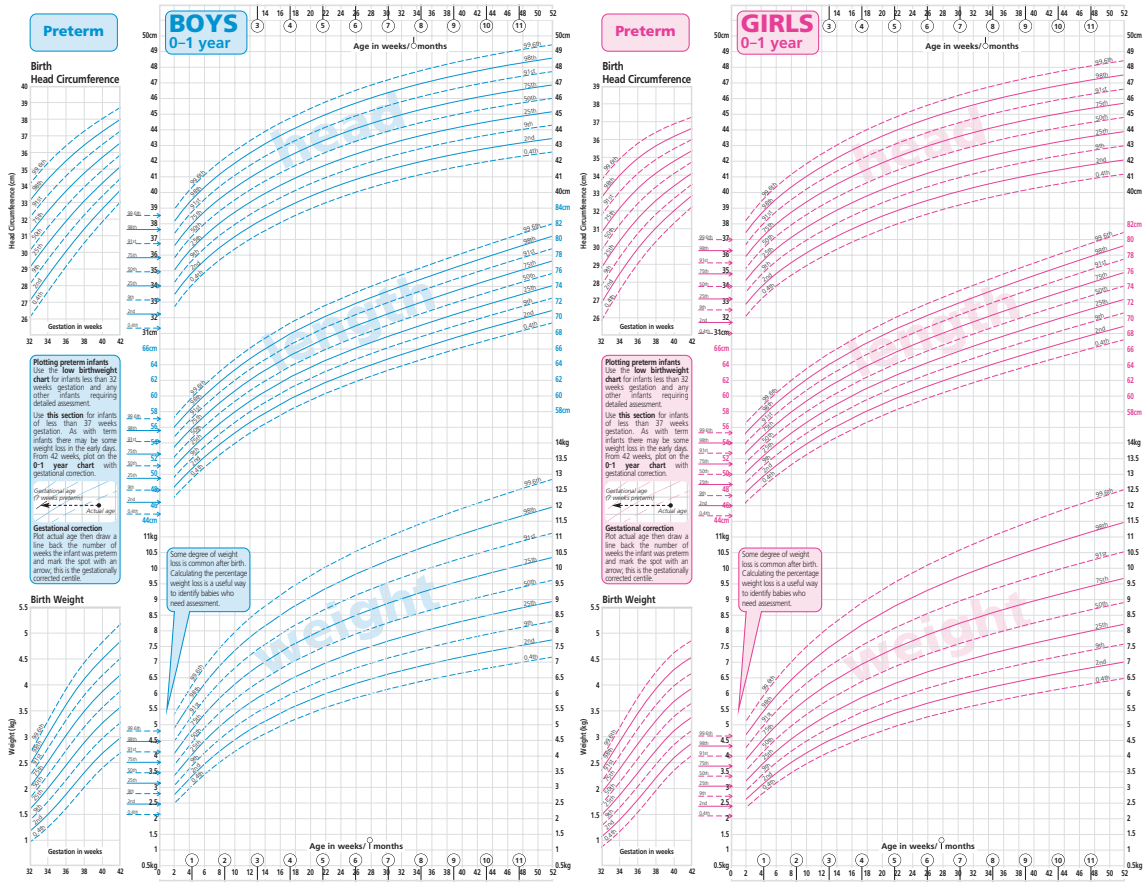


Figure A.1, cont'd Growth charts 0–1 year Boys (A.1c) Girls (A.1d). (RCPCH/WHO/Department of Health. ©2009 Department of Health).

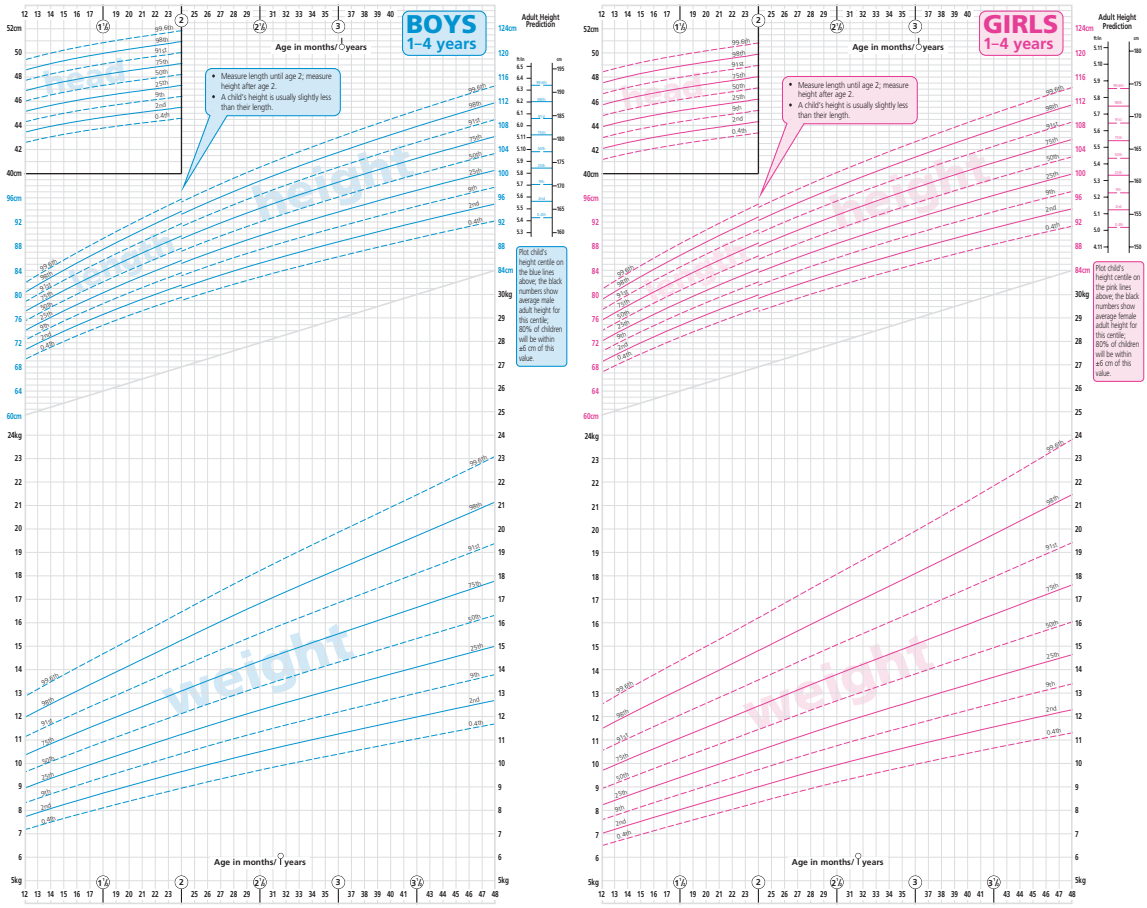


Figure A.1, cont'd Growth charts 1-4 years Boys (A.1e) Girls (A.1f). (RCPCH/WHO/Department of Health. ©2009 Department of Health.)

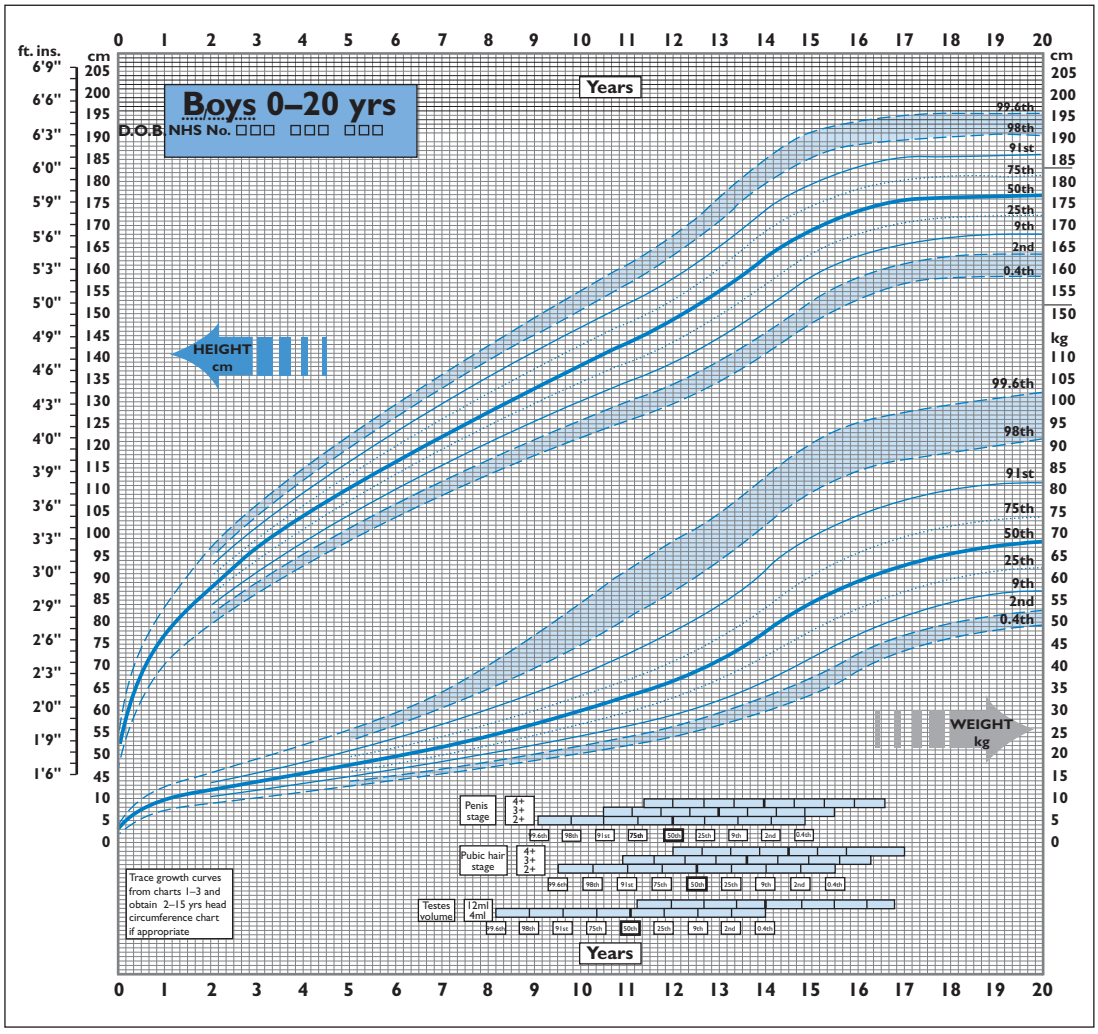


Figure A.1, cont'd Growth chart of males from birth to 20 years using the nine centile UK chart (A.1g). The interval between each pair of centile lines is the same (⅔ standard deviation). It shows the 0.4 and 99.6 centile lines, which are 2⅔ standard deviations below and above the median respectively. It also shows the timing of the stages of puberty. (Chart ©Child Growth Foundation.)

Gestational age assessment of newborn infants

Neuromuscular maturity

	–1	0	1	2	3	4	5
Posture							
Square window (wrist)	>90°	90°	60°	45°	35°	0°	
Arm recoil		180°	140°–180°	110°–140°	90°–110°	<90°	
Popliteal angle	180°	160°	140°	120°	100°	90°	<90°
Scarf sign							
Heel to ear							

Physical maturity

Skin	Sticky, friable, transparent	Gelatinous, red, translucent	Smooth pink, visible veins	Superficial peeling and/or rash, few veins	Cracking pale areas rare veins	Parchment, deep cracking, no vessels	Leathery, cracked, wrinkled
Lanugo	None	Sparse	Abundant	Thinning	Bald areas	Mostly bald	
Plantar surface	Heel-toe 40–50 mm: –1 < 40 mm: –2	>50 mm no crease	Faint red marks	Anterior transverse crease only	Creases ant. 2/3	Creases over entire sole	
Breast	Imperceptible	Barely perceptible	Flat areola, no bud	Stippled areola 1–2 mm bud	Raised areola 3–4 mm bud	Full areola 5–10 mm bud	
Eye/ear	Lids fused loosely: –1 tightly –2	Lids open Pinna flat, stays folded	Sl. curved pinna; soft; slow recoil	Well-curved pinna; soft but ready recoil	Formed and firm, instant recoil	Thick cartilage, ear stiff	
Genitalia male	Scrotum flat, smooth	Scrotum empty, faint rugae	Testes in upper canal, rare rugae	Testes decending, few rugae	Testes down, good rugae	Testes pendulous, deep rugae	
Genitalia female	Clitoris prominent, labia flat	Prominent clitoris, small labia minora	Prominent clitoris, enlarging minora	Majora and minora equally prominent	Majora large, minora small	Majora cover clitoris and minora	

(c) Maturity rating

Score	Weeks
–10	20
–5	22
0	24
5	26
10	28
15	30
20	32
25	34
30	36
35	38
40	40
45	42
50	44

Figure A.2 Scoring system for assessment of gestational age in newborn infants (Ballard score). This is a method of assessing gestational age according to neuromuscular and physical maturity. The infant's gestation or age (±2 weeks) is determined from the total score using the conversion chart. (Adapted from: Ballard JL, Khoury JC, Wedig K , et al. 1991. New Ballard score, expanded to include extremely premature infants. *Journal of Pediatrics* 119:417–423.)

Management plan for asthma

Green zone – GO

Your asthma is under control if:
- your breathing feels good
- you do not have a cough or wheeze
- you can take part in normal activities and play games / sport.
- your are sleeping through the night
- you are not missing school because of your asthma

Peak Flows are between:

……

……

Green zone action –
Your normal medicines are:
Preventer

……

……

Other Medicines

……

……

Reliever

……
as required.
Remember: If necessary take this before exercise or if you have cold-like symptoms. Take 2–4 puffs every 4 hours if you need it.
If there is no improvement, move to the Amber zone.

Amber zone – WARNING

Your asthma is getting worse if you:
- wake at night with asthma symptoms
- have a cough, wheeze or 'tight' chest
- need to use the reliever inhaler once a day, more than usual or it is not lasting for four hours.

Peak Flows are between:

……

……

Amber zone action –
Take all medicines as normal.

Take 4–10 puffs of reliever – one puff at a time as taught. Use a spacer if you have one and shake the MDI in between puffs. Take every four hours if needed.

- If no improvement – make an appointment to see your doctor for that day.
- If you have a Symptom/Peak Flow Diary – start filling it in and take it with you to the doctor.

Or
Start your home prednisolone …….mgs daily, if you have it. See your doctor if you are not better after 12 hours.

Remember: If no better after 10 puffs of reliever, move to Red zone.

Red zone – DANGER

Your asthma is severe if after taking 10 puffs of reliever you:
- are still breathing hard and fast
- can't talk or feed easily
- are exhausted
- are frightened and look anxious
- are very pale/grey/blue in colour

Peak Flow reading (if able) below:

……

……

Red zone action –
Call an ambulance now

- Keep taking one puff of reliever every 20–30 seconds or 4 slow breaths or, if you have one, nebulized reliever with oxygen until the ambulance arrives.
- Take a dose of oral steroids if not already taken.
- Don't move about
- Keep calm

Figure A.3 Example of patient management plan for asthma.

Blood pressure chart

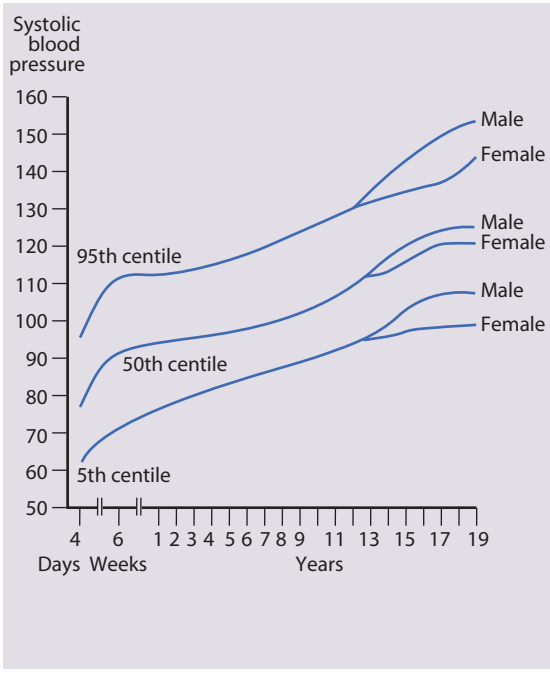


Figure A.4 Systolic blood pressure according to age. Blood pressure charts are also available according to height. (Data from: de Swiet M, Fayers P, Shinebourne EA. 1980. Blood pressure in a population of infants in the first year of life: the Brompton Study. *Pediatrics* 65:1028–1035 and de Man SA, Andre JL, Bachmann H et al. 1991. Blood pressure in childhood: pooled findings of six European studies. *Journal of Hypertension* 9:109–114.)

Peak flow chart

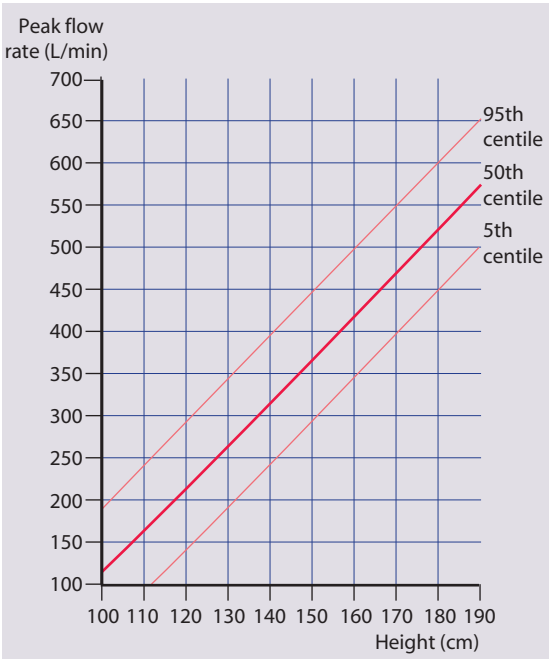


Figure A.5 The normal range of peak flow measurements according to height. (Reproduced with permission from: Godfrey S, Kamburoff PL, Nairn JR. 1970. Spirometry, lung volumes and airway resistance in normal children aged 5 to 18 years. *British Journal of Diseases of the Chest* 64:15–24.)

Normal ranges: haematology

Age	Hb (g/dl)	MCV (fL)	WBC ($\times 10^9$/L)	Platelets ($\times 10^9$/L)
Birth	14.5–21.5	100–135	10–26	150–450 at all ages
2 weeks	13.4–19.8	88–120	6–21	
2 months	9.4–13.0	84–105	6–18	
1 year	11.3–14.1	71–85	6–17.5	
2–6 years	11.5–13.5	75–87	5–17	
6–12 years	11.5–15.5	77–95	4.5–14.5	
12–18 years:				
Male	13.0–16.0	78–95	4.5–13	
Female	12.0–16.0	78–95	4.5–13	

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